

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

The History of the Bar Code

A

The first step toward today's bar codes came in 1948, when Bernard Silver, a graduate student in the USA, overheard a conversation in the halls of Philadelphia's Drexel Institute of Technology. The president of a food chain was pleading with a professor to undertake research on a method of capturing product information automatically at store checkouts. The professor turned down the request, but Bernard Silver mentioned the conversation to his friend Norman Woodland, a twenty-seven-year-old teacher at Drexel.

B

The problem fascinated the two friends, and they set about thinking of a solution. Their first idea was to use patterns printed with an ink that would glow under ultraviolet light, and they built a device to test the concept. It worked, but the printing costs were high and the patterns faded over time. Nonetheless, they were convinced they had a workable idea. After several months of work, they came up with the linear bar code, using elements from two established technologies: Morse code, in which letters and numbers are coded into a system of dots and dashes, and the method used to record soundtracks in movies. Silver and Woodland patented the idea in 1952, describing their invention as 'article classification... through the medium of identifying patterns'. But the cost, together with the fact that their scanning equipment was rather unreliable, made the idea a non-starter at that time.

C

Scanning systems made little progress until the 1970s, when lasers became affordable. Following this, various systems came into use around the world in stores, libraries, factories, and the like, each with its own proprietary code, but there was no standardisation. A consortium of grocery manufacturers and retailers therefore set up a committee to look into bar codes and to standardise what became known formally as the Universal Product Code (UPC). At the heart of the committee's guidelines were a few basic principles. To make life easier for the cashier, bar codes would have to be readable from almost any angle and at a range of distances. Because they would be reproduced by the million, the labels would have to be cheap and easy to print. And to be affordable, automated checkout systems would have to pay for themselves in two and a half years.

D

The committee considered more than a dozen versions of bar codes, including one based on multi-coloured dots and another using a circular bull's-eye design with lines radiating from a central point. On April 1, 1973, they unanimously agreed on a standardised UPC: a combination of black and white lines and numbers, based on Woodland and Silver's idea but developed by George Laurer at IBM. Alan Haberman, who headed the subcommittee as

president of First National Stores, described the bar code as a kind of world language that worked for everyone. He recalls proudly, 'We showed that it could be done on a massive scale, that cooperation was possible for the common good, and that business didn't need the government to shove them in the right direction.'

E

The investment involved in the bar-code revolution was huge. Each of the tens of thousands of grocery outlets in the US had to spend at least \$200,000 on new scanning equipment. Chains had to install new data-processing centres and retrain their employees. Printers had to develop new types of ink, plates, and other technology to reproduce the code with the exact tolerances it requires, and manufacturers had to spend millions of dollars a year on the labels.

F

On June 26, 1974, all the tests were done, all the proposals were complete, and all the standards were set. At a supermarket in Ohio, a single pack of Juicy Fruit chewing gum became the first retail product sold with the help of a bar-code scanner. Decades of schemes and billions of dollars in investment now became a practical reality. The bar code on any product could be read and understood in every suitably equipped store.

G

The advantages of the system were not clear immediately, as wholesalers, retailers and customers remained suspicious. Some customers believed bar codes were a form of surveillance. During the early weeks, *Business Week* magazine ran the headline 'The Supermarket Scanner That Failed'. However, the benefits eventually became apparent. 'It turns out there were massive savings in labour and other areas,' Haberman says. These included checking out items at twice the speed compared to using traditional equipment, which meant shorter queues. It did not take supermarkets long to see that, as well as vastly improving customer service, the bar code could hugely reduce the time spent checking inventory.

H

Now, every day more than five billion bar codes are scanned in retail outlets throughout the world. Passengers' luggage is tagged with bar codes by airlines. Staff attach them to babies to ensure the right babies go home from hospitals with the right mothers. Runners in major marathons set off with bar codes on their vests, and librarians rely on them. Tiny bar codes have even been mounted on bees by researchers to track their movements.

I

As for that original pack of Juicy Fruit, it is now, unchewed and unopened, in the Smithsonian Institution's National Museum of American History in Washington.

Questions 1–8

Complete the notes below.

Write **ONE WORD ONLY** from the passage for each answer.

History of the Bar Code

1948–1952

Methods of recording information automatically were developed by Silver and Woodland.

First system:

- used ultraviolet light and a special type of **1**
- problems: expensive and not permanent

Second system:

- based on technology used in Morse code and also for the **2** of films
- problems: **3** and expensive

1970s

Availability of cheaper **4** meant scanning technology spread more widely.

Problem: lack of **5** in code systems.

April 1973: committee agreed on one Universal Product Code (UPC)

June 1974: pack of Juicy Fruit chewing gum sold with bar-code scanner

Advantages of bar-code system:

- supermarkets needed to spend less on labour
- the **6** of checkouts increased
- doing inventories was much cheaper

Present day

Users of bar codes include:

- retail companies
- airlines
- staff in hospitals
- participants in **7**
- scientists studying **8**

Questions 9–13

TRUE

if the statement agrees with the information given in the text

FALSE

if the statement contradicts the information

NOT GIVEN

if there is no information on this.

9 Bernard Silver was invited to develop a system for capturing product information by the president of a food chain.

10 A committee set up in the 1970s said bar codes should be easy to use and not too expensive.

11 Alan Haberman disagreed with government policies on business matters.

12 Many grocery outlets were unable to afford the necessary scanning equipment.

13 The advantages of the new bar-code scanner took some time to be accepted by users.

READING PASSAGE 2

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2 below.

Should We Stop Eating Meat?

A

A call for people to eat less meat to save the planet is growing louder, so if less is good, wouldn't none be better? To find out, let's imagine what would happen if the world eliminated meat, milk and eggs from its diet, then trace the effects. Last year, the world consumed 289 million tonnes of meat, 700 million tonnes of milk and 1.2 billion eggs. Environmentally speaking, this came at an enormous cost.

B

All agriculture damages the environment. Take, for example, the felling of forests and the widespread use of irrigation systems. It may surprise you to know that agriculture creates more greenhouse gases than all methods of transport put together. Livestock farming does the most damage. In part, that is because most livestock eat grain that could be used to feed human populations, and farmers are forced to grow more than we would otherwise need to meet the demand. Altogether, if we switched to a vegan diet, meaning no meat, dairy or eggs, the land currently required for crops would drop by an estimated 21 per cent, about 3.4 million hectares, roughly the size of India.

C

One environmental impact that would also lessen through a reduction in animal farming would be nitrogen emitted from agricultural processes, which spreads into both waterways and land. According to environmental scientist Allison Leach, if everyone eliminated dairy products and eggs, this kind of pollution would fall by 60 per cent. Livestock production has another serious environmental impact. Global statistics are hard to come by, but in the US at least, livestock account for 55 per cent of erosion, mostly from forests being cut down to make way for grazing land. On top of this, half of all antibiotics manufactured are fed to livestock as part of their normal diet, a practice that is leading to drug-resistant bacteria.

D

A meat-free world, then, would be greener in many ways. However, if everyone opted to give up meat, there would be significant costs too. For most of human history, livestock grazed on land that wasn't suitable for ploughing, and in doing so they converted inedible grass into edible meat and milk. Even today, a flock of sheep or goats can be the most efficient way to get food from marginal land. In a world in which over a billion people do not have enough to eat, using this land for crop production would contribute to food insecurity.

E

In some parts of the world today, livestock such as chickens can subsist on leftovers and whatever they find. Tara Garnett, who heads the Food Climate Research Network at the University of Surrey, points out the usefulness of these animals. By giving them your leftovers, she says, they deal with your rubbish, and you get meat. Garnett admits, though, that if this kind of approach were generally adopted, it would require a major adjustment in food preferences; people would have to get used to chicken, for example, with less fat.

F

Another downside to a meat-free world would be the disappearance of animal by-products. Such a world would have to replace the 11 million tonnes of leather and 2 million tonnes of wool that come annually from livestock farming and which are turned into clothing. Furthermore, even ardent vegetarians acknowledge that dairy products and meat may be a good thing in poorer countries. 'Whilst there's no doubt that considerable reduction of meat consumption would have an environmental benefit, we do have to be careful about saying it would be the best solution if the whole world went vegetarian,' says Annette Pinner, chief executive of the UK Vegetarian Society. For many of the world's poorest rural residents, an animal may represent their only realistic hope for a little extra income, and a little animal protein can make a big difference to a marginal diet.

G

What if we decided on a no-meat vegetarian diet, rather than a vegan diet? After all, milk and eggs are very efficient ways of producing animal calories. 'It's difficult to switch to a no-meat-but-milk diet,' says Helmut Haberl, a social ecologist. 'Dairy cows must calve every year to keep producing milk, and only half their offspring will be female. While many vegetarians see moral reasons not to kill and eat the males, there is surely no practical reason to waste so much meat.' Similar arguments apply to chickens kept for eggs.

H

So even though a meat-free world sounds good on paper, it is likely that a utopian future will still have some animal products in it. The real questions, then, are how much meat do we want, and how will we produce it? The answers depend on how you approach these questions. The most straightforward approach is to assume that the world will continue to want ever more meat. The United Nations' best guess is that by 2050 the world will need to more than double its production of meat, an increase that would be environmentally disastrous.

I

Under this scenario, the goal will have to be producing the most meat at the lowest environmental cost. According to Walter Falcon, an agricultural economist, this means fewer free-range cattle and sheep in green fields. 'If you're going to keep some livestock systems, I think the ones you'll want to keep are the intensive ones,' he says. Of course, this does not take into account animal welfare issues, as intensive farming usually means poor living conditions and the use of growth hormones.

Questions 14–17

*Reading Passage 2 has **nine paragraphs, A–I.***

Which paragraph contains the following information?

*Choose the correct letter, **A–I.***

- 14** the suggestion that people may need to adapt to a different quality of meat
- 15** a description of the way animals were fed in the past
- 16** a prediction in regard to human demand for meat
- 17** the potential consequences of a meat-free world for textile industries

Questions 18–22

Complete the summary below.

*Write **ONE WORD ONLY** from the passage for each answer.*

How does agriculture damage the environment?

Agriculture causes significant harm to the environment in a number of ways, including the frequent use of **18** in modern farming and the cutting down of trees. It is also responsible for the production of many greenhouse gases, more so than **19** is. However, it is livestock farming in particular that is most harmful, because cows, chickens and sheep, for example, are given vast quantities of grain which could be consumed by people instead. A meat-free diet would mean that actually less **20** would be needed. A meat-free diet would also mean less chemical pollution, specifically in the form of **21**, especially if consumers cut out dairy and egg products as well as meat. In the US, there is clear evidence that considerable **22** occurs, although there is no information on this from other countries.

Questions 23–26

Look at the following statements (Questions 23–26) and the list of people below.

*Match each statement with the correct person, **A–E**.*

*Write the correct letter, **A–E**, in boxes 23–26 on your answer sheet.*

List of People

- A** Allison Leach
- B** Tara Garnett
- C** Annette Pinner
- D** Helmut Haberl
- E** Walter Falcon

- 23** It is not possible to say that a vegetarian diet is right for everyone.
- 24** It may be economically preferable to farm animals in limited space.
- 25** It does not make sense to give up meat without giving up dairy products too.
- 26** Some animals can be fed in a way that allows waste to be recycled.

READING PASSAGE

You should spend about 20 minutes on Questions 27–40, which are based on the passage below.

Robert Louis Stevenson

A

It is more than 100 years since the death of the Scottish writer Robert Louis Stevenson on the South Pacific island of Samoa, and it seems that time has not been kind to Stevenson's memory. Immediately after his death, his family and friends set to work to fashion the legend of Robert Louis Stevenson, or RLS, as he became known — one of the few writers familiar from his initials alone. Subsequent works of biography then turned him into a writer of almost religious importance. One example was the literary critic Balfour, who in 1901 portrayed Stevenson's family as ministering angels to the dying genius during his final illness. Similarly, the biographer Crouch absurdly overstated Stevenson's significance by placing him in the same company as the most revered names in English literature: Shakespeare and Keats. The reaction to this nonsense was a number of highly critical assessments of Stevenson's legacy in the 1920s.

B

Normally, the critical pendulum can be relied on to swing back again, but there are several aspects of Stevenson's work that have, until recently, acted against a more balanced appraisal. First is the allegation that Stevenson was a mere master of linguistic fireworks who lacked moral depth. Some critics accused him of being a literary charlatan, of juggling words very prettily to strike effects which overawed an ignorant public and served to distract from the inadequacy of his ideas.

C

Then there has long been a prejudice against the adventure story as the proper medium for deep moral seriousness, a prejudice which is still extremely influential today. It seems that we can accept that an adventure film can successfully express profound moral truths, but we reject the same idea for a book. The absurdity of this becomes apparent when we think of writers like Joseph Conrad and Graham Greene, but it is no use pretending that this bias against adventure stories is not part of our high culture. A further problem is that Stevenson has often not found favour in the land of his birth because his conservatism so often collides with the strong radical tradition in Scotland. His many escapist stories and preference for living abroad have led to accusations that he camouflaged Scotland's social problems. Lastly, the high adventure of Stevenson's own lifestyle has sometimes obscured his output. His globe-trotting, and above all the final phase of his life in Samoa, tended to make his own life a greater story than any he could devise. This was precisely what his friends feared would happen towards the end of his short life: that his art might be overwhelmed by the drama of life in Samoa.

D

One consequence of this has been that Stevenson's influence on other writers has too often been neglected. The writer and poet Oscar Wilde was deeply influenced by Stevenson, even though he declared that Stevenson would have produced better work if he had lived in London rather than Samoa. Stevenson tends to stick in the throat even of those writers who would like to spit him out, such as Shaw, who claimed to have learned from him that the romantic hero is always mocked by reality. Likewise, the writer Galsworthy, who began as a determined critic, later changed his mind and said that the superiority of Stevenson over the novelist Hardy was that Stevenson was all life and Hardy all death. The influence on the novelist Chesterton would also repay detailed study, for it was through him that Stevenson has managed to cross the ages, emerging as an influence on the modernist movement and the contemporary Latin American school of 'magical realism'.

E

When making an assessment of his life and work, one question must inevitably be asked: was Robert Louis Stevenson Scotland's greatest writer of English prose? For most commentators, this honour falls to Sir Walter Scott, author of *Ivanhoe* among many other classic novels, and it is true that in terms of craftsmanship, precision and the ability to minutely regulate language to create the desired effect, Scott takes the prize. However, this is not the same thing at all as inherent talent. By way of comparison, one may take the example of the two great Russian composers Shostakovich and Prokofiev, of whom the former had learned more precise skills of execution but the latter's intrinsic genius was greater — and so it seems to be with Scott and Stevenson. Admittedly, Scott's detailed style does permit his stories to explore levels of tragedy that are beyond Stevenson's reach, but in this regard they have the musty smell of the museum, somehow artificial and removed from modern-day reality. On the other hand, Stevenson's skill with plotting and narrative gives his books a timeless quality, so that they still live today. Stevenson was also the shrewder judge of behaviour and psychology. For example, his compelling descriptions of a man with a split personality in *The Strange Case of Dr Jekyll and Mr Hyde* have proved so accessible and accurate that the expression 'Jekyll and Hyde' has entered common English usage. Even if we do not see a revival of critical interest in this great Scottish writer, it is to be hoped that readers return to Robert Louis Stevenson's magnificent stories and reassess this neglected genius.

Questions 27–31

Choose the correct answer, **A**, **B**, **C** or **D**.

27 In the opinion of the writer, the biographers Balfour and Crouch

- A** understated the role played by Stevenson's family
- B** misunderstood Stevenson's religious beliefs
- C** overestimated other writers' influence on Stevenson
- D** elevated Stevenson above his true status as a writer

28 What point does the writer make about Stevenson in the second paragraph?

- A** The public judged him more fairly than the critics.
- B** Recent criticism of him has been justified.
- C** Critics argued that his style covered up his faults.
- D** The ethical nature of his stories was often criticised.

29 According to the writer, the adventure story

- A** is more appropriate for books than films
- B** can be used by writers to tell moral stories
- C** is more fashionable today than in the past
- D** has been used by other writers but not Stevenson

30 What point does the writer make about Stevenson and Scotland?

- A** His ideas contrasted with those of many Scots.
- B** He demonstrated great sympathy for Scotland's problems.
- C** He was not considered a true Scot as he was not born there.
- D** His unflattering stories about Scotland angered many Scots.

31 According to the writer, Stevenson's own lifestyle

- A** was envied by his friends
- B** was responsible for his early death
- C** attracted more attention than his books
- D** did not prepare him for living in Samoa

Questions 32–35

Do the following statements agree with the claims of the writer?

Write **YES**, **NO**, or **NOT GIVEN**.

32 Although Oscar Wilde admired Stevenson's work, he believed Stevenson could have written something better.

33 Stevenson encouraged Oscar Wilde to start writing.

34 Galsworthy had greater respect for Hardy than Stevenson.

35 More research is needed regarding Stevenson's influence on Chesterton.

Questions 36–40

Complete the summary using the list of words, **A–I**, below.

Robert Louis Stevenson and Sir Walter Scott

Opinions differ as to whether Robert Louis Stevenson or Sir Walter Scott should be considered Scotland's best writer. Scott had greater **36**, but Stevenson had more **37**, and the same distinction can be made between the two composers Shostakovich and Prokofiev. It is true that Scott's books showed more **38** when it came to tragedy, though in an old-fashioned way, while Stevenson's books are still popular because of his **39** Stevenson's understanding of **40** has also resulted in the widespread use of an expression from one of his books.

Word Bank

A natural ability
B critical acclaim
C humour

D romance
E colourful language
F technical control

G storytelling
H depth
I human nature

Answers and Explanations

PASSAGE 1

Questions 1–8: Notes Completion

1. ink

Explanation: Paragraph B describes their first idea as using patterns printed with an ink that glows under ultraviolet light.

2. soundtracks

Explanation: Paragraph B states they used elements from “the method used to record soundtracks in movies.”

3. unreliable

Explanation: Paragraph B notes that their scanning equipment was “rather unreliable,” which was a problem.

4. lasers

Explanation: Paragraph C says scanning systems progressed in the 1970s “when lasers became affordable.”

5. standardisation

Explanation: Paragraph C explains that before the UPC, there was “no standardisation” among different systems.

6. speed

Explanation: Paragraph G mentions that bar codes allowed “checking out items at twice the speed.”

7. marathons

Explanation: Paragraph H notes that “runners in major marathons” use bar codes on their vests.

8. bees

Explanation: Paragraph H states that researchers have mounted bar codes on bees to track their movements.

Questions 9–13: True/False/Not Given

9. FALSE

Explanation: Paragraph A states the president asked a professor, who declined. Silver overheard the conversation but was not directly invited.

10. TRUE

Explanation: Paragraph C outlines the committee's principles, including that bar codes should be readable from any angle (easy to use) and that systems should pay for themselves quickly (affordable).

11. NOT GIVEN

Explanation: Paragraph D quotes Haberman saying business didn't need government direction, but it does not state whether he disagreed with specific government policies.

12. NOT GIVEN

Explanation: Paragraph E mentions the cost of equipment but does not state that many outlets were unable to afford it.

13. TRUE

Explanation: Paragraph G explains that advantages "were not clear immediately" and that suspicions existed before benefits became apparent.

PASSAGE 2:

Answers and Explanations

Questions 14–17: Paragraph Matching

14. E

Explanation: Paragraph E mentions that if people ate chickens fed on leftovers, they would have to "get used to chicken... with less fat."

15. D

Explanation: Paragraph D describes how historically, livestock grazed on land unsuitable for crops.

16. H

Explanation: Paragraph H states the UN prediction that meat production will more than double by 2050.

17. F

Explanation: Paragraph F discusses how a meat-free world would mean replacing leather and wool from livestock.

Questions 18–22: Summary Completion

18. irrigation

Explanation: Paragraph B mentions “widespread use of irrigation systems.”

19. transport

Explanation: Paragraph B says agriculture creates more greenhouse gases than “all methods of transport put together.”

20. land

Explanation: Paragraph B says land required for crops would drop by 21% if the world went vegan.

21. nitrogen

Explanation: Paragraph C discusses nitrogen emitted from farming as a pollutant.

22. erosion

Explanation: Paragraph C states livestock account for 55% of erosion in the US.

Questions 23–26: Matching Statements to People

23. C – Annette Pinner

Explanation: In Paragraph F, Pinner says we must be careful claiming a vegetarian diet is best for the whole world.

24. E – Walter Falcon

Explanation: In Paragraph I, Falcon argues intensive livestock systems (limited space) have the lowest environmental cost.

25. D – Helmut Haberl

Explanation: In Paragraph G, Haberl says it’s impractical to give up meat but keep dairy, as male calves would be wasted.

26. B – Tara Garnett

Explanation: In Paragraph E, Garnett explains chickens can be fed leftovers, recycling waste into meat.

PASSAGE 3

Answers and Explanations

Questions 27–31: Multiple Choice

27. D – elevated Stevenson above his true status as a writer

Explanation: Paragraph A states that Balfour and Crouch exaggerated Stevenson's importance, comparing him to Shakespeare and Keats.

28. C – Critics argued that his style covered up his faults.

Explanation: Paragraph B describes critics accusing Stevenson of using linguistic tricks to distract from the inadequacy of his ideas.

29. B – can be used by writers to tell moral stories

Explanation: Paragraph C argues that adventure stories, like those by Conrad and Greene, can convey profound moral truths.

30. A – His ideas contrasted with those of many Scots.

Explanation: Paragraph C explains that Stevenson's conservatism clashed with Scotland's radical tradition.

31. C – attracted more attention than his books

Explanation: Paragraph C notes that Stevenson's adventurous lifestyle often overshadowed his literary work.

Questions 32–35: Yes/No/Not Given

32. YES

Explanation: Paragraph D states Oscar Wilde believed Stevenson would have written better if he had lived in London instead of Samoa.

33. NOT GIVEN

Explanation: The passage mentions Wilde was influenced by Stevenson but does not say Stevenson encouraged him to start writing.

34. NO

Explanation: Paragraph D says Galsworthy eventually considered Stevenson superior to Hardy.

35. YES

Explanation: Paragraph D suggests that Stevenson's influence on Chesterton "would repay detailed study," implying more research is needed.

Questions 36–40: Summary Completion

36. F – technical control

Explanation: Paragraph E describes Scott's superior craftsmanship, precision, and language regulation.

37. A – natural ability

Explanation: Paragraph E contrasts Scott's technical skill with Stevenson's greater inherent talent or genius.

38. H – depth

Explanation: Paragraph E states Scott's stories explore levels of tragedy beyond Stevenson's reach, indicating greater depth.

39. G – storytelling

Explanation: Paragraph E credits Stevenson's skill with plotting and narrative for his books' timeless quality.

40. I – human nature

Explanation: Paragraph E highlights Stevenson's shrewd understanding of behavior and psychology, as seen in Dr Jekyll and Mr Hyde.